ANATOMY 3 NEUROANATOMY MOCK 3

1- How many pairs of spinal nerves are there?

A- 8
B- 33
C- 31
D- 12

2- Which cells produce myelin in the CNS?

A- Oligodendrocytes
B- Astrocytes
C- Microglia
D- Schwann cells

3- From which germ layer is the nervous system derived?

A- Endoderm
B- Ectoderm
C- Neural crest only
D- Mesoderm

4- What is the foramen of Pacchioni?

A- The opening bounded by the free border of the tentorium
B- The median aperture of the fourth ventricle
C- The opening between lateral and third ventricles
D- The opening of the aqueduct

5- Which lobe is the largest, about 40% of the hemisphere?

A- Temporal
B- Parietal
C- Occipital
D- Frontal

6- Which commissure passes under the pineal gland, in the posterior wall of the third ventricle?

A- Habenular commissure
B- Posterior white commissure
C- Corpus callosum
D- Anterior white commissure

7- Through which structure do the basal ganglia send their output back to the cortex?

A- Cerebellum
B- Thalamus
C- Hippocampus
D- Spinal cord

8- Epilepsy as a sign of a hemispheric lesion points to:

A- A cortical lesion
B- A peripheral nerve lesion
C- A cerebellar lesion
D- A spinal cord lesion

9- What connects the two thalami?

A- Corpus callosum
B- Interthalamic adhesion
C- Fornix
D- Posterior commissure

10- What is the most posterior part of the thalamus?

A- Habenula
B- Mammillary body
C- Pulvinar
D- Interthalamic adhesion

11- The hypothalamus is the interface between:

A- The nervous and endocrine systems
B- The two hemispheres
C- The cerebrum and cerebellum
D- The spinal cord and the muscles

12- What is the lateral boundary of the subthalamus?

A- Hypothalamus
B- Third ventricle
C- Internal capsule
D- Red nucleus

13- On which bone does the brainstem lie?

A- Basilar part of the occipital bone (clivus)
B- Squamous occipital bone
C- Body of the sphenoid only
D- Petrous temporal bone

14- Which cranial nerve emerges from the dorsal surface of the brainstem?

A- Abducens (VI)
B- Oculomotor (III)
C- Trochlear (IV)
D- Trigeminal (V)

15- What marks the border between pons and medulla on the floor of the fourth ventricle?

A- Medial eminence
B- Facial colliculus
C- Stria medullaris
D- Obex

16- Which ascending pathway is unconscious?

A- Lateral spinothalamic tract
B- Dorsal column-medial lemniscus
C- Spinocerebellar tracts
D- Anterolateral system

17- Why is tonsillar herniation life-threatening?

A- It compresses the medulla
B- It tears the falx
C- It blocks the optic chiasm
D- It compresses the cauda equina

18- Which is the largest deep cerebellar nucleus?

A- Fastigial
B- Emboliform
C- Dentate
D- Globose

19- The cerebrocerebellum sends its output through:

A- The fastigial nucleus
B- The vestibular nuclei
C- The dentate nucleus
D- The inferior olive

20- Why does the adult spinal cord end above the end of the vertebral canal?

A- The cord shrinks with age
B- The vertebral column keeps growing after the cord stops
C- The cauda equina pulls it up
D- The cord is cut at birth

21- What anchors the spinal cord to the coccyx?

A- Cauda equina
B- Filum terminale
C- Denticulate ligament
D- Conus medullaris

22- What is the denticulate ligament?

A- The end of the cord
B- Lateral extensions of the pia attaching to the dura
C- A ligament of the vertebral bodies
D- A fold of dura

23- Which basal ganglion lies in the midbrain?

A- Subthalamic nucleus
B- Putamen
C- Substantia nigra
D- Caudate nucleus

24- What joins the hypothalamus to the pituitary?

A- The cerebral aqueduct
B- The habenula
C- The fornix
D- The infundibulum

25- Which deep nucleus is linked to balance and posture through vestibular connections?

A- Emboliform
B- Fastigial
C- Dentate
D- Globose

ANSWER KEY

1- C
2- A
3- B
4- A
5- D
6- B
7- B
8- A
9- B
10- C
11- A
12- C
13- A
14- C
15- C
16- C
17- A
18- C
19- C
20- B
21- B
22- B
23- C
24- D
25- B